

CODE SPORT



In this month's column, we discuss a medley of computer science interview questions.

We are at the last column of the year and it seems like 2021 has gone by in a whirl. We have been discussing foundational models in AI in our last columns. However, since it is our practice to share computer science interview questions in the year-end column, we will take a break from our discussion of AI foundation models and discuss a few ML related interview questions.

1. You are given the problem of disambiguating entities in a given set of documents. For example, for the word 'Jaguar', you need to disambiguate between the use of the word as a luxury car and as a ferocious animal, based on the context. A more complicated example is the following:
S1: The White House announced its intention to support the newly formed government in Haiti.
S2: The New Year party will be held at the White House, and will be hosted by Jill Biden.
In sentence S1, White House is an entity that represents the American government. In sentence S2, White House is a location. Can you come up with an algorithm that can disambiguate such entity mentions to their appropriate types based on the context? This task is known as named entity disambiguation. A related but broader problem is the task of word sense disambiguation (WSD), where the word can be anything and not just an entity. For example, the word 'bank' can refer to the bank of a river as well as a commercial entity. Though these problems are related problems, WSD is considered easier to solve, since the contexts for each of the senses associated with the word are typically very different. This is not the case with entity type disambiguation.
2. You have been asked to build an application that analyses the conversations in call centres. These conversations typically happen between the agent and the customer. An automatic speech recognition (ASR) system has been used to convert the voice

conversations into text. It is possible that it may have mis-recognised some of the words and hence the transcript may occasionally contain wrong words instead of the actually spoken words. You are asked to give a method that can go through these transcripts automatically, and (a) identify which of the words in the transcript are incorrect; (b) with user confirmation, correct each wrong word with the appropriate correct word.

Your first thought is to use a standard dictionary to identify words in the transcript which are not in the dictionary, and mark them as wrong words. But this has two sources of errors: (a) A domain specific word may not always be in a general dictionary. Hence, just the absence of a word in a dictionary does not make it an incorrect word; (b) the transcript may contain a word that is actually in the dictionary, but it could be a word recognised wrongly by the ASR system. How will you handle both these types of issues?

3. You are asked to build a system that can do part of speech (POS) tagging for a given sentence. While POS tagging assigns independent labels for each of the words in the sentence, it uses the sequence of words it has seen in the context to decide on the current label. For example, consider the sentence: "I saw the saw on the table and took it outside to the shed." The first occurrence of 'saw' in the above sentence is a verb whereas the second occurrence of the word 'saw' is a noun. By looking at the sequence of words occurring around the specific word, a POS tagging system can distinguish between the tags it should assign for each of the two occurrences of the word 'saw'. Hence, POS tagging is an example of sequence labelling. Can you come up with standard algorithms for sequence labelling?
4. We are all familiar with different sorting and searching algorithms. However, these algorithms